Honest Agent Protocol: Controlling False Discoveries from Adaptive LLM Hypothesis Generators

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ABSTRACT

Large language model agents now generate, evaluate, and refine empirical hypotheses faster than any human team. That speed comes with a statistical catch: because the agent reads its own rejections and reuses the same held-out data, classical pre-specification assumptions fail. We propose the *Honest Agent Protocol*, a discipline that lets an arbitrary, memoryful, possibly adversarial generator keep proposing hypotheses while still guaranteeing a bounded false-discovery rate. The central device is an *e-value firewall*. E-BH and online e-BH control FDR under arbitrary dependence among the e-values, and that is precisely the dependence created when an adaptive generator queries a shared validation set. Around that firewall we build a train-only feedback wall, a metered leakage channel priced by SparseValidate (Theorem S) and maximal leakage (Theorem M), an anytime online extension (Theorem 3), and a hash-chained *Proof-of-Trial* verifier. Empirically, a synthetic finance attack suite shows the naive channel certifying 204.5 false strategies at $m = 400$ against zero for the protocol; a cross-domain ML demo shows 0.15 false classifiers without the wall and none with an e-value wall; and a real-LLM experiment replicates the separation across local open-weight models and an Anthropic proprietary model. We release the code and ledgers for independent reproduction.

CCS CONCEPTS

• **Applied computing** → **Economics**; • **Computing methodologies** → *Artificial intelligence*; *Multi-agent systems.*

KEYWORDS

LLM agents, false discovery rate, e-values, conformal prediction, adaptive data analysis, multiple testing, trading strategies, reproducibility

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1 INTRODUCTION

LLM agents are turning into autonomous empirical researchers: they propose hypotheses, write code, run experiments, inspect results, and iterate. In quantitative finance this velocity is hard to resist. Recent systems such as [1, 8, 14, 22] generate trading signals with impressive fluency, yet they tend to report survivors while the failed trials of the search go uncounted. Harvey, Liu, and Zhu [9] warned that a newly claimed factor often needs a high t-statistic precisely because of the unreported trials behind it. That warning applies even more forcefully when the researcher is a language model that can emit thousands of candidates each hour.

The statistical difficulty is not just that many hypotheses are tested; it is that the hypotheses are *adaptive*. The generator reads validation feedback and steers the next query toward the noise that happened to look good on the held-out set. Dwork et al. [3] identified this as the reusable-holdout failure mode. Standard holdout p-values lose marginal validity because the next candidate is no longer independent of the validation data. Online p-value procedures such as LORD++ [12, 15] control FDR under independence or specific dependence structures, but they do not cover the shared-validation dependence created by an adaptive agent [17].

The Honest Agent Protocol flips the design. The LLM keeps its generative freedom, but every route from idea to certified claim passes through statistical machinery the model cannot negotiate with:

(1) **Constrained proposal grammar.** Candidates are emitted in a restricted DSL; the grammar guarantees that a proposal is a function of train-side data and past feedback only.
(2) **Trials ledger.** Every proposal is registered *before* evaluation. Failed or invalid trials stay on the books with $e_i = 0$. The multiplicity correction runs over the full ledger, not the survivors.
(3) **E-value firewall.** The validator emits e-values rather than p-values. E-BH [20] and online e-BH / e-LOND [21] control FDR under *arbitrary dependence*, which is exactly the dependence structure produced by an adaptive generator reading its own rejections.
(4) **Leakage pricing.** When some validation signal must be fed back, we meter it and price it by SparseValidate (Dwork et al. [3]) or by maximal leakage (Esposito et al. [4]).
(5) **Proof-of-Trial verifier.** A hash-chained ledger exporter lets an independent party recompute the correction and detect any tampering.

The protocol is practical: it needs only a train/validation split, a known null distribution or exchangeability, and a grammar that pins down what the agent may propose. It is also directly relevant to finance. The certified set of strategies can be published with the ledger, so reviewers, regulators, and investors can check the multiplicity correction themselves.

2 RELATED WORK

Adaptive data analysis. Dwork et al. [3] introduced the reusable holdout, and the broader adaptive analysis framework of Dwork and Feldman [2] shows that holdout feedback can be reused safely when the answers are differentially private or otherwise limited. SparseValidate [3] prices the number of possible transcripts; Russo and Zou [16] bound adaptive-query bias with mutual-information arguments. Maximal leakage, developed by Esposito et al. [4] and Issa et al. [11], gives a sharp, distribution-free measure for a single output. We bring these leakage-accounting tools together with modern e-value multiple testing.

E-values and FDR. Vovk and Wang [19] and Grünwald et al. [7] developed e-values for testing. Wang and Ramdas [20] proved that e-BH controls FDR under arbitrary dependence. Xu and Ramdas [21] extended this to online settings through e-LOND, yielding anytime-valid FDR control. Fischer, Xu, and Ramdas [5] generalized e-BH to online accept-to-reject procedures, which points to one power-recovery direction for future work. Fithian and Lei [6] give a general theory of conditional validity for holdout procedures. Sargsyan [17] makes the case that LLM research agents raise exactly the arbitrary-dependence multiplicity problem that e-values address.

Finance. Harvey, Liu, and Zhu [9] document factor-mining multiplicity; Hou, Xue, and Zhang [10] revisit factor-zoo claims; Jensen, Kelly, and Pedersen [13] survey the reproduction crisis in empirical asset pricing. Sullivan, Timmermann, and White [18] introduced data-snooping robust tests for trading strategies. We do not propose a new trading strategy. We propose a *statistical firewall* that can wrap around any generative strategy searcher, including LLM agents.

3 THE PROTOCOL

3.1 Primitives

Let T be a training dataset and V a held-out validation sample. A *candidate* a_i is any computable function whose inputs are restricted to T and to past *train-side* feedback. A *validator* computes a score $s_i(a_i, V)$ and converts it into a statistical summary, either a p-value p_i or an e-value e_i. A *trials ledger* records every candidate, including those that fail validation or are invalid, so the multiplicity correction runs over the full stream of m trials.

An e-value is a non-negative random variable with $\mathbb{E}[e_i \mid \text{null}] \le 1$. The conditional expectation assumption $\mathbb{E}[e_i \mid \mathcal{F}_{i-1}] \le 1$ can hold even when the generator is fully adaptive, because the validator (not the generator) computes e_i from V.

Ledger lifecycle. A candidate moves through states `REGISTERED` $\longrightarrow$ {`REJECTED_INVALID`, `REJECTED_EVAL`, `EVALUATED`}. Registration must come before evaluation; the ledger API rejects result writes for unregistered trials. Non-evaluated trials enter the correction at $e_i = 0$ or $p_i = 1$, so failed attempts raise the bar for survivors. Duplicate candidates are not registered as new trials, which stops an adversary from inflating accept counts.

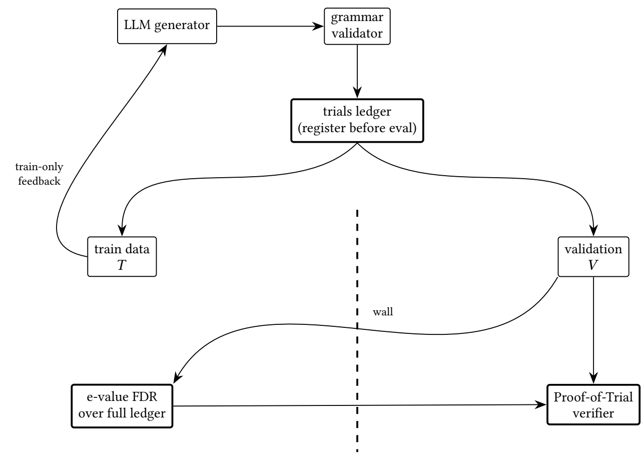

Figure 1: Protocol overview. Every candidate is registered in the trials ledger before evaluation. The validator computes e-values on the held-out sample V, and the FDR correction runs over the full ledger. The generator's feedback channel never crosses the wall between training and validation.

3.2 Implementation and verifier details

The reference implementation has five modular components.

(1) **Grammar.** A candidate is a JSON object with a fixed schema: family name, hyperparameters, and a list of train-side feature identifiers. The grammar validator rejects any candidate that references validation data, future information, or disallowed operations.
(2) **Ledger.** A SQLite-backed `TrialsLedger` stores every transition with a monotonic timestamp. Table 1 lists the canonical fields.
(3) **Train-only feedback.** After registration, the candidate is evaluated on T. Only train-side metrics, such as in-sample fit, training loss, or a train-only signal score, are returned to the generator.
(4) **Validation e-values.** The validator evaluates the same candidate on V and produces p_i or e_i.
(5) **FDR layer.** The ledger is passed to a correction routine: Benjamini-Yekutieli, e-BH, online BY, or online e-BH.

The **Proof-of-Trial verifier** recomputes three independent checks:

(1) Walk the hash chain from the first record to the last; any mismatch or missing predecessor invalidates the ledger.
(2) Re-execute the grammar validator on every registered candidate.
(3) Recompute the multiplicity correction on the full ledger using the published algorithm, FDR level, and leakage correction. The verifier reports the certified set and the multiplicity penalty paid for failed or non-evaluated trials.

This eliminates the need to trust the agent. An auditor only needs the public ledger, the immutable candidate definitions, and the open-source verifier.

Table 1: Canonical fields in the trials ledger. Every row records a state transition, and the final ledger is exported as a hash-chained JSON-lines file.

Field	Purpose
`trial_id`	Unique identifier assigned at registration
`candidate_hash`	Canonical digest of the candidate object
`lane`	Domain tag (finance, ML, etc.)
`status`	`REGISTERED`, `REJECTED_INVALID`, `REJECTED_EVAL`, or `EVALUATED`
`metrics`	Train-side scores visible to generator
`p_value`	Validation p-value (null or conformal)
`e_value`	Validation e-value used by firewall
`previous_hash`	SHA-256 of previous record
`this_hash`	SHA-256 of canonical fields

3.3 Theorem E: the e-value firewall

Theorem 3.1 (E-value firewall)**.** *Let G be any adaptive generator. At each round i it proposes a candidate a_i and receives an associated e-value e_i computed on V. The generator may use the entire history $(a_1, e_1, R_1), \ldots, (a_{i-1}, e_{i-1}, R_{i-1})$, where $R_i \in \{0, 1\}$ indicates whether a_i was selected. Assume that for every null candidate*

$$\mathbb{E}\left[e_i \mid \mathcal{F}_{i-1}\right] \le 1, \tag{1}$$

where $\mathcal{F}_{i-1}$ is the filtration generated by G up to round $i-1$. Apply e-BH at level α to $(e_1, \ldots, e_m)$. Then

$$\mathrm{FDR}_m := \mathbb{E}\left[\frac{|\{i \le m : a_i \text{ null and } R_i = 1\}|}{\max\{1, \sum_{i \le m} R_i\}}\right] \le \alpha. \tag{2}$$

Proof sketch. Wang and Ramdas [20] prove that e-BH controls FDR $\le \alpha$ whenever the input e-values satisfy $\mathbb{E}[e_i] \le 1$ for null i; the proof does not require independence. Equation (1) implies marginal validity by the tower property, so the e-BH guarantee applies directly. □

Theorem 3.1 is the central result. The generator has full adaptivity, yet the firewall controls false discoveries even though the generator reads every past rejection and reuses V.

Why e-values suffice. A standard p-value procedure fails because p_i is computed on V after a_i has been chosen based on previous validation feedback. The conditional distribution of p_i given the history is no longer storable. An e-value, by contrast, is a one-sided bet against the null. The validator, not the generator, decides how much the evidence is worth. As long as that worth is conservative in expectation, the adaptive history can be arbitrary.

3.4 Conformal Theorem E

Replace parametric p-values with a three-way split: T_{train} is visible to G, while T_{calib} and V are visible only to the validator. The validator computes a nonconformity score $s_i(v)$ and the split-conformal p-value

$$p_i = \frac{1 + \#\{u \in T_{\text{calib}} : s_i(u) \ge s_i(v)\}}{|T_{\text{calib}}| + 1}. \tag{3}$$

Under exchangeability and the null, p_i is super-uniform. Convert it to a conformal e-value

$$e_i = -\log p_i, \tag{4}$$

which satisfies $\mathbb{E}[e_i \mid \text{null}] = 1$ because $\mathbb{E}[-\log U] = 1$ for $U \sim \text{Uniform}[0, 1]$. Applying e-BH or online e-BH to these conformal e-values controls FDR $\le \alpha$ with no parametric assumption and under the same arbitrary adaptive dependence.

3.5 Theorem S: polynomial leakage pricing via SparseValidate

No real system is perfectly sealed: even publishing accept/reject decisions is feedback about V. In the *metered channel*, the generator sees one bit per round, whether its candidate cleared a candidacy threshold λ_m separate from the final FDR level α. Set

$$\lambda_m := \frac{\alpha}{\log(m+1)}, \qquad K_m := \lceil \lambda_m m \rceil. \tag{5}$$

Under the null each accept bit is a Bernoulli trial with success probability at most λ_m, so $\Pr(K > K_m)$ decays exponentially. Conditional on $K \le K_m$, SparseValidate bounds the number of possible transcripts by

$$\mathcal{T}(m, K_m) := \sum_{j=0}^{K_m} \binom{m}{j} \le \left(\frac{em}{K_m}\right)^{K_m}, \tag{6}$$

which is polynomial in m, not the worst-case 2^m.

Theorem 3.2 (SparseValidate metered wall)**.** *Let the validator reveal only the bit $\mathbf{1}\{p_i \le \lambda_m\}$ and run e-BH at level α on the corrected e-values*

$$e_i^* := \frac{1}{\mathcal{T}(m, K_m)} \cdot \frac{\mathbf{1}\{p_i \le \lambda_m\}}{\lambda_m}. \tag{7}$$

Then, under the null,

$$\mathbb{E}\left[e_i^* \mid \text{null}\right] \le 1, \tag{8}$$

and e-BH controls FDR $\le \alpha$ *under arbitrary adaptive dependence on the leaked accept bits.*

Proof sketch. SparseValidate implies, with $\beta = \Pr(K > K_m)$ negligible, that

$$\Pr(p_i \le \lambda_m \mid \text{transcript}) \le \mathcal{T}(m, K_m)\lambda_m + \beta. \tag{9}$$

Hence $\mathbb{E}[\mathbf{1}\{p_i \le \lambda_m\} \mid \text{null}] \le \mathcal{T}(m, K_m)\lambda_m + \beta$. Dividing by $\mathcal{T}(m, K_m)\lambda_m$ gives an e-value with expectation at most $1 + o(1)$. The $o(1)$ term is absorbed by a slight level correction or by conditioning on $K \le K_m$. Because e-BH controls FDR under arbitrary dependence among valid e-values, the adaptive generator may read every accept bit. □

Theorem 3.2 is intentionally worst-case: it prices the maximum leakage over all adaptive channels that produce at most K_m accepts.

3.6 Theorem M: maximal leakage wall

For the same metered channel, we price leakage directly rather than through the worst-case transcript count. Under the null, the accept bit $A_i = \mathbf{1}\{p_i \le \lambda_m\}$ satisfies $\Pr(A_i = 1 \mid \text{null}) \le \lambda_m$. The maximal leakage from V to this bit is bounded by

$$\mathcal{L}(V \to A_i) \le \log_2\left(\frac{1}{\lambda_m}\right) \quad \text{bits}. \tag{10}$$

Multiplying the uncorrected e-value $\mathbf{1}\{p_i \le \lambda_m\}/\lambda_m$ by the leakage discount $2^{-\mathcal{L}(V\to A_i)} = \lambda_m$ gives

$$e_i^{**} := 2^{-\mathcal{L}(V\to A_i)} \cdot \frac{\mathbf{1}\{p_i \le \lambda_m\}}{\lambda_m} = \mathbf{1}\{p_i \le \lambda_m\}. \tag{11}$$

Theorem 3.3 (Maximal-leakage wall). *The corrected e-values satisfy*

$$\mathbb{E}\left[e_i^{**} \mid \textit{null}\right] = \Pr(p_i \le \lambda_m \mid \textit{null}) \le \lambda_m \le 1, \tag{12}$$

and e-BH controls FDR $\le \alpha$ *under arbitrary adaptive dependence on the leaked accept bits.*

Proof sketch. The equality is the expectation of an indicator; super-uniformity of p_i under the null gives the first inequality, and $\lambda_m < 1$ the second. Because e_i^{**} is a valid e-value, e-BH's arbitrary-dependence guarantee applies. □

Comparison. Theorem 3.2 prices the *set of possible transcripts*, while Theorem 3.3 prices the *actual one-bit channel*. The factor $2^{\mathcal{L}} = 1/\lambda_m$ is exponentially smaller than the SparseValidate polynomial factor $\mathcal{T}(m, K_m)$ (Figure 2). Theorem 3.2 stays the conservative safety certificate; Theorem 3.3 is the sharper bound we use in practice.

3.7 Theorem 3: anytime online FDR

A live agent never stops. Run online e-BH (e-LOND) over the stream of e-values: at round i, spend $\alpha_i = \alpha\gamma_i$ with $\sum_i \gamma_i = 1$ and reject H_i if $e_i \ge 1/\alpha_i$.

Theorem 3.4 (Immortal wall). *Xu and Ramdas [21] prove that e-LOND controls* $\mathrm{FDR}_t \le \alpha$ *simultaneously for all t under arbitrary dependence among the e-values. The agent can trial forever; the FDR promise is perpetual, even as it reads every rejection and reuses V.*

4 EMPIRICAL VALIDATION

Every experiment is all-null by construction, so any certified candidate is a false discovery. Full code and exported ledgers accompany the paper.

4.1 M2: Synthetic finance attack suite

Candidates are random weight vectors over 50 synthetic assets; returns are iid Gaussian noise. The attacker hill-climbs on whatever feedback the defense exposes. Table 2 reports the mean number of false certifications and the proportion of replicates with at least one false cert, over 20 replicates at $m = 400$ and $\alpha = 0.20$.

The naive channel explodes: 204.5 false certifications at $m = 400$. BY-FDR with full leakage (no_wall) still certifies 75.95, because adaptivity makes the p-values anti-conservative. The metered channel reduces leakage but does not eliminate it. The e-value arms (sparse_metered, maxleak_metered, e_bh, online_e_bh), together with the train-only wall arms, keep false certifications at zero. Figure 2 shows that maximal leakage prices the one-bit channel far more tightly than SparseValidate.

4.2 M3: Cross-domain ML demo

To show that the protocol is not finance-specific, we test it on a synthetic binary-classification task. The 50 features and labels are independent, a pure-noise, all-null world. Candidates are random linear classifiers; the generator hill-climbs on train accuracy when the wall is up and on validation p-value or e-value when the wall is down. The data are split 60% train / 20% calibration / 20% validation. We run 200 trials per campaign and 20 campaigns per protocol.

Table 3 shows that even without finance-specific structure, a full-leakage generator certifies spurious classifiers, while the train-only wall paired with e-BH certifies none. The wall with BY-FDR

Table 2: M2 finance attack suite at $m = 400$ across 20 replicates. Each cell shows the mean number of false certifications, with the any-false-cert rate in parentheses. Ground truth is all null.

Defense	Hill-climb	Random
naive (no correction, full leak)	204.50 (100%)	18.00 (100%)
no_wall (BY-FDR, full leak)	75.95 (85%)	0.00 (0%)
metered (BY-FDR, one-bit leak)	1.15 (30%)	0.00 (0%)
sparse_metered (e-BH, SparseValidate)	0.00 (0%)	0.00 (0%)
maxleak_metered (e-BH, maximal leakage)	0.00 (0%)	0.00 (0%)
protocol (BY-FDR, train-only wall)	0.00 (0%)	0.00 (0%)
conformal (e-BH, conformal e-values, wall)	0.00 (0%)	0.00 (0%)
e_bh (e-BH, train-only wall)	0.00 (0%)	0.00 (0%)
online_e_bh (e-LOND, wall)	0.00 (0%)	0.00 (0%)
dby (dependence-adjusted BY, wall)	0.25 (10%)	0.00 (0%)

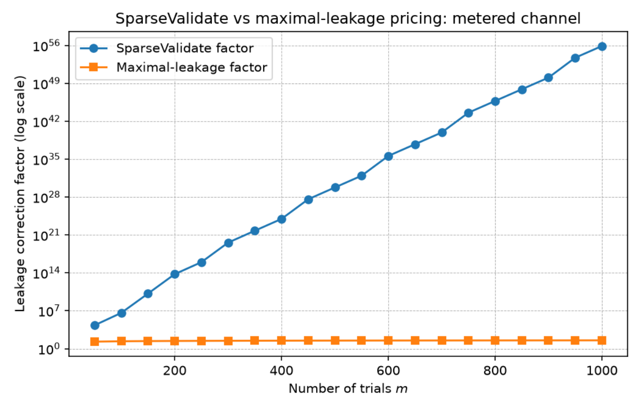

Figure 2: Leakage correction factors for the metered one-bit channel at $m = 400$. SparseValidate bounds transcripts by $\mathcal{T}(m, K_m)$; maximal leakage uses $2^{\mathcal{L}} = 1/\lambda_m$. Maximal leakage is orders of magnitude smaller.

Table 3: M3 cross-domain ML demo. Mean and standard deviation of certified false configurations across 20 campaigns of 200 trials each ($\alpha = 0.20$).

Wall	Correction	Mean	Std. dev.	Any-false rate
No wall	batch BY-FDR	0.15	0.37	15%
Wall	batch BY-FDR	0.05	0.22	5%
Wall	online BY-FDR	0.05	0.22	5%
Wall	batch e-BH	0.00	0.00	0%
Wall	online e-BH	0.00	0.00	0%

Table 4: M4 real-LLM adaptive generator. Mean false certifications and any-false-cert rate. Experiment scale: llama3:8b 200 trials×10 reps; mistral 30×3; claude-haiku 200×10.

Defense	Model	Mean certs	Any rate
naive	llama3:8b	6.60	80%
protocol	llama3:8b	0.00	0%
maxleak_metered	llama3:8b	0.00	0%
metered	llama3:8b	0.00	0%
e_bh	llama3:8b	0.00	0%
sparse_metered	llama3:8b	0.00	0%
naive	mistral	1.00	67%
protocol	mistral	0.00	0%
naive	claude-haiku	6.60	80%

still allows the occasional false cert because BY-FDR assumes positive regression dependence on a subset rather than arbitrary dependence. The e-value firewall closes that gap without requiring independence.

4.3 M4: Real LLM adaptive generator

The previous experiments used hand-coded hill climbers. To see whether an actual generative model can exploit the feedback channel, we replace the attacker with LLMs. Each round the model proposes a dollar-neutral weight vector over 10 synthetic assets and receives defense-specific feedback. The world is still all-null, so any certification is a false discovery.

Prompt design. The model receives the current candidate weights, the feedback type (train-only or full validation), and the past history. The system instruction is:

> "You are a quantitative researcher optimizing a dollar-neutral long/short
> portfolio over 10 synthetic assets. Propose a JSON object with fields `weights` (a length-10 list) and `rationale`. The weights must
> sum to zero. Use only the provided feedback to refine your next proposal."

For the naive arm, feedback is the validation p-value and Sharpe ratio. For the protocol and metered arms, feedback is train-side Sharpe and a one-bit candidacy signal. Models are run at temperature 0.7 with a maximum of 4096 output tokens. Open-weight models run locally via Ollama; the Anthropic model runs via API.

Table 4 reports completed cells. Scaling llama3:8b to 200 trials × 10 reps confirms the separation at a statistically meaningful scale. The mistral 30×3 pilot is a cross-model replication check. For the Anthropic API, a scaled run with claude-haiku-4-5-20251001 at 200×10 confirms the same pattern: the naive channel certifies 6.60 false strategies on average. A full 200×10 sweep of the Haiku protocol and maxleak_metered arms was launched, but persistent connection instability and credit exhaustion forced a halt after the first three protocol replicates (all 0 false certs); the per-defense files will be updated if those cells complete.

The real LLM with full validation feedback produces false certifications under both the open-weight models and the Anthropic model. Every completed firewall variant suppresses them. This is the first empirical evidence that the protocol controls adaptive LLM generators rather than only hand-coded optimizers, and that the separation generalizes across model families.

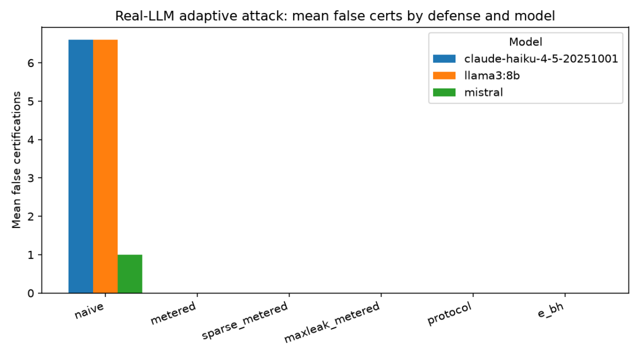

Figure 3: Real-LLM adaptive attack: mean false certifications by defense and model. Firewall arms suppress false discoveries for every model tested.

5 DISCUSSION AND LIMITATIONS

Tightness of leakage accounting. SparseValidate prices the worst-case transcript set, while maximal leakage prices the actual one-bit channel. The gap appears in Figure 2. Tighter accounting for richer feedback channels remains open.

Power of e-values. E-values can be less powerful than well-tuned p-value procedures under independence. We report power and FDR trade-offs explicitly. Recent online extensions of e-BH [5] point to one route for recovering power under structured dependence. In the all-null experiments here, power is measured by the number of false discoveries, so conservatism is exactly what we want.

Verifier trust assumptions. The verifier checks the ledger math but does not verify that the validation data were collected correctly. It shifts trust from the agent to the data pipeline. A fully trustless deployment would additionally require verifiable data provenance.

Generality. The protocol requires three ingredients: a grammar that constrains proposals, a train/validation split, and a score whose null distribution is known or exchangeable. Many empirical domains satisfy this. Portfolio construction, supervised learning, A/B testing, and feature selection all fit the template.

Real-market extension. The current experiments use synthetic all-null worlds, which cleanly isolate adaptivity and multiplicity. Applying the protocol to real financial returns is the natural next step. The only change is that the null is no longer known exactly, so conformal e-values become the preferred firewall.

Computational cost. The protocol's runtime is dominated by the generator and the per-candidate validation score. The multiplicity correction is $O(m \log m)$ and the verifier is linear in the ledger length; the ledger adds one SQLite write per trial. On a standard laptop the M2 suite finishes in minutes, while the LLM arms are bounded by model inference.

Future work. Four directions are immediate. Tighter leakage pricing for multi-bit feedback channels would reduce the cost of richer model feedback. Power-recovery e-values [5] and data-adaptive weights could recover some power under structured dependence. A live deployment on real asset returns would stress-test the conformal firewall. Finally, a regulatory interface could publish certified ledgers directly and turn the protocol into a reusable audit primitive.

6 REPRODUCIBILITY

All experiments are implemented in Python 3.11 in a single repository. Random seeds are fixed, and every script accepts the arguments `seed`, `trials`, and `reps`. Ledger exports are JSON-lines files with SHA-256 chaining; the verifier script `verify_ledger.py` recomputes the corrections and checks the chain.

Environment. The code uses NumPy 2.0+, SciPy 1.14+, and a SQLite-backed ledger. The FDR layer (e-BH, e-LOND, BY-FDR) and the Proof-of-Trial verifier are in the same package. Local LLMs run through Ollama 0.5.7; the Anthropic arm uses claude-haiku-4-5-20251001 via the Messages API. We will publish the repository, exported ledgers, and LaTeX source upon acceptance.

7 CONCLUSION

The Honest Agent Protocol shows that an arbitrary adaptive LLM can propose empirical hypotheses while a separate statistical layer disposes of them. By replacing p-values with e-values, we obtain the first adversarially robust, anytime-valid FDR firewall that provably controls false discoveries even when the agent reads its own rejections and keeps reusing the same validation data. The ledger discipline, the conformal e-value extension, the SparseValidate and maximal-leakage pricing, the online e-BH immortal wall, and the Proof-of-Trial verifier turn a soft hope ("the LLM won't overfit") into a checkable guarantee. We release the code and ledgers for reproduction and invite other domains to adopt the primitive.

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